

CLASS 10 · CBSE · NCERT MATHEMATICS · DPP SHEET

Chapter 8: Introduction to Trigonometry

Daily practice problems: 10 questions with an answer key

Try every question on your own first. The answer key is on the last page.

1. In $\triangle ABC$, right-angled at B , $AB = 24$ cm and $BC = 7$ cm. Find $\sin A$ and $\cos A$.
2. If $\cos A = \frac{4}{5}$, find $\sin A$ and $\tan A$.
3. If $\sec \theta = \frac{13}{12}$, find $\sin \theta$ and $\cot \theta$.
4. Evaluate $\sin^2 30^\circ + \cos^2 60^\circ + \tan^2 45^\circ$.
5. Evaluate $3 \tan^2 30^\circ + 4 \cos^2 45^\circ - \sec^2 60^\circ$.
6. Evaluate $\frac{\sin 60^\circ + \cos 30^\circ}{1 + \tan 45^\circ}$.
7. If $\sin(A + B) = \frac{\sqrt{3}}{2}$ and $\tan(A - B) = \frac{1}{\sqrt{3}}$, with $0^\circ < A + B \leq 90^\circ$ and $A > B$, find A and B .
8. Prove that $(1 + \cot^2 A)(1 - \cos A)(1 + \cos A) = 1$.
9. Prove that $\cos^4 A - \sin^4 A = 1 - 2 \sin^2 A$.
10. Prove that $(\sec A - \cos A)(\cot A + \tan A) = \tan A \sec A$.

Answer Key

1. $AC = 25 \text{ cm}$; $\sin A = \frac{7}{25}$, $\cos A = \frac{24}{25}$

2. $\sin A = \frac{3}{5}$, $\tan A = \frac{3}{4}$

3. $\sin \theta = \frac{5}{13}$, $\cot \theta = \frac{12}{5}$

4. $\frac{1}{4} + \frac{1}{4} + 1 = \frac{3}{2}$

5. $1 + 2 - 4 = -1$

6. $\frac{\sqrt{3}}{2}$

7. $A + B = 60^\circ$, $A - B = 30^\circ$: $A = 45^\circ$, $B = 15^\circ$

8. $\text{LHS} = \operatorname{cosec}^2 A \times \sin^2 A = 1$

9. $\text{LHS} = (\cos^2 A + \sin^2 A)(\cos^2 A - \sin^2 A) = \cos^2 A - \sin^2 A = 1 - 2 \sin^2 A$

10. $\sec A - \cos A = \frac{\sin^2 A}{\cos A}$ and $\cot A + \tan A = \frac{1}{\sin A \cos A}$; the product is $\frac{\sin A}{\cos^2 A} = \frac{\sin A}{\cos^2 A}$